AUTHENTISCRIBE

Product Requirements Document: AI Integrity & Linguistic Intelligence Workspace

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Author	Product Team
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1. Overview

AuthentiScribe is a full-stack writing workspace built for people who need to check whether text is original, and if it's not, actually fix it. It's aimed at educators, researchers, and professional writers who currently juggle separate tools for plagiarism checking, AI detection, rewriting, and citations. This product tries to put all of that in one place.

Under the hood, it's a React and TypeScript frontend talking to an Express backend, with Google's Gemini API doing the heavy lifting for text analysis, rewriting, and structured output.

2. Problem Statement

Since generative AI tools became mainstream, checking whether a piece of writing is genuinely human has turned into a real headache for a lot of people. The tools that exist today solve pieces of the problem but not the whole thing. You run a plagiarism check in one app, an AI detector in another, then copy the flagged paragraph into a third tool to rewrite it, and maybe a fourth to get the citation format right. None of these tools talk to each other.

That back and forth is where AuthentiScribe is trying to help. Instead of detecting a problem and leaving the user to go fix it elsewhere, it keeps detection and remediation in the same screen so a flagged sentence can be reviewed, rewritten, and checked again without switching tabs.

3. Goals & Objectives

- Bring AI detection, plagiarism analysis, rewriting, and citation formatting together in one workspace instead of several disconnected tools.
- Show the reasoning behind a flag, not just a score. Perplexity, burstiness, and sentence-level classification should be visible, not hidden behind a single number.
- Make it possible to act on a flag right away through the Originality Bench and the Humanizer, rather than just reporting the issue and stopping there.
- Work with the files people already have. Native PDF and DOCX support matters more than a text-paste-only tool.

- Keep it free and open source (Apache-2.0) so individuals and smaller institutions can self-host it without a licensing cost.

4. Who This Is For

- **Educators.** They need to check student work for AI-generated content and plagiarism, and they need evidence they can actually point to, not just a percentage.
- **Researchers and academic writers.** They want to catch AI-pattern language in a draft before submission, and get citations formatted correctly without hunting through a style guide.
- **Professional and content writers.** They often start with an AI-assisted draft and need to turn it into something that reads naturally and matches their voice, while also cleaning up grammar.
- **Students.** They want to understand where their own writing might get flagged and reduce that risk before they turn something in.

5. Scope

5.1 What's Included

- AI-generation and plagiarism detection for both pasted text and uploaded files (PDF, DOCX)
- Sentence-level charts showing perplexity, burstiness, and how each sentence is classified
- A highlight viewer that color-codes human, AI-generated, and plagiarized text
- A humanizer with adjustable rewrite intensity and style presets
- Grammar checking, summarization, and translation tools
- Reference finding and citation formatting in APA, MLA, and Chicago styles
- Session history and export of cleaned documents or reports

5.2 What's Not Included (for now)

- User accounts, authentication, or billing for a hosted SaaS version
- Persistent server-side storage of documents beyond the current session
- Native mobile apps for iOS or Android
- Real-time collaboration between multiple users on the same document
- Direct integrations with LMS platforms like Canvas or Google Classroom

6. Features

These are grouped the way they already exist in the product, roughly by workspace tab.

6.1 Plagiarism & AI Detection Suite

Feature	Description	Priority
Multi-engine analysis	Audits pasted text or uploaded documents and estimates AI-generation likelihood alongside plagiarism probability.	High
Linguistic integrity charting	Sentence-by-sentence charts (built with Recharts) plotting perplexity, burstiness, and category	High

Feature	Description	Priority
	classification.	
Interactive highlight viewer	Color-codes human, AI-generated, and plagiarized text as you read. Clicking a flagged sentence opens the Originality Bench.	High

6.2 Humanizer & Writing Module

Feature	Description	Priority
Context-aware rephrasing	Rewrites stiff, machine-sounding language into more natural prose while keeping the original meaning intact.	High
Adjustable profiles	Choose rewrite intensity (Low, Medium, High) and a style preset such as Academic, Business, Casual, or Narrative.	Medium
Interactive suggestions	Offers a few different humanized versions of a sentence or paragraph so the user can pick the one that fits best.	Medium

6.3 File Handling

Feature	Description	Priority
Direct PDF processing	PDFs are base64-encoded and sent straight to the Gemini API, which parses them natively.	High
DOCX extraction	Word files are parsed server-side using mammoth.js.	High
Embedded document preview	A split-screen canvas layout lets you inspect the uploaded PDF locally while working.	Medium

6.4 Editorial Toolkit

Feature	Description	Priority
AI proofreader & translator	Checks structure and style, and translates content into other languages.	Medium
Reference finder & citation assistant	Finds likely source material and formats citations in APA, MLA, or Chicago style.	Medium
History & export	Lets you revisit earlier documents from a local session panel, and export cleaned text or a full report.	Low

7. Non-Functional Requirements

- Analysis should come back within a few seconds for a typical document. This depends partly on how fast the Gemini API responds, which is outside our direct control.
- Gemini's responses need to follow a strict JSON schema so the frontend can render results without guessing at the shape of the data.

- API keys stay server-side, stored as environment variables, and are never sent to the browser.
- The interface should support both dark and light themes and hold up on different screen sizes.
- The whole thing should run as a single Node.js app that anyone can install and self-host with one build and start command.
- Documents are processed per session only. Nothing gets saved permanently on the server unless a future version adds that.

8. Technical Architecture

8.1 Stack

- Frontend: React 18, TypeScript, Vite
- Styling: Tailwind CSS
- Charts: Recharts, used for the perplexity and burstiness visualizations
- Animation: Framer Motion
- Backend: Express.js on Node with TypeScript
- AI: Google Gemini API through the official @google/genai SDK
- Document parsing: mammoth.js for DOCX files

8.2 How Data Moves Through the App

A user either pastes text directly or uploads a file. If it's a DOCX, the server pulls the raw text out using mammoth.js. If it's a PDF, the file is base64-encoded and handed to Gemini, which reads it natively without needing a separate extraction step. Either way, the resulting text goes to the analysis endpoint as a JSON payload, Gemini returns a structured response, and the dashboard renders that response as charts and highlighted text. If the user clicks into a flagged sentence and asks for a rewrite, that request goes to a separate endpoint that also calls Gemini, just with a different prompt.

8.3 API Endpoints

- POST /api/extract-text: pulls raw text out of an uploaded DOCX file
- POST /api/analyze: runs the AI-detection and plagiarism analysis through Gemini
- POST /api/rephrase-sentence: generates humanized rewrites for a flagged sentence or paragraph

9. Core User Flow

- User pastes text, or uploads a PDF or DOCX file.
- The file gets parsed, either server-side for DOCX or natively by Gemini for PDF.
- The analysis engine reviews the text and sends back a structured integrity report.
- The dashboard shows sentence-level highlights alongside the stylometric charts.
- User clicks a flagged sentence, which opens the Originality Bench.
- User picks a rewrite intensity and style, and the Humanizer suggests alternatives.
- Once the user is happy with the result, they can add citations through the Reference Finder and export the cleaned document or a full report.

10. Success Metrics

These are early proposals. Some will need real usage data before we can set a meaningful target.

Metric	Target	How we'd measure it
Detection accuracy	Low false-positive rate against known human-written samples	Spot-checking against a control set of verified human writing
Time to fix a flagged sentence	Shorter time between a flag appearing and an accepted rewrite	Timing from highlight click to accepted suggestion within a session
Adoption	Steady growth in stars and self-hosted deployments	GitHub repository analytics
Completion rate	A healthy share of uploaded documents reach export or report stage	Session funnel tracking, once instrumentation exists

11. Assumptions & Constraints

- Users bring their own Gemini API key. The product itself doesn't sell or resell API access.
- How well analysis performs, and what it costs, depends on Gemini's availability, rate limits, and pricing at any given time.
- Right now this is a single-tenant, self-hosted tool. There's no built-in multi-user account system.
- Document history lives locally in the session. It isn't backed by a central database yet.

12. Risks & Mitigations

Risk	Impact	Mitigation
False positives or negatives in AI detection	High, especially for educators who rely on this as evidence	Frame results as probabilistic evidence with sentence-level detail, not a final verdict, and encourage a human to make the final call.
Everything depends on one AI provider (Gemini)	Medium, an outage or pricing change hits core functionality directly	Build the AI-calling layer so it can support other model providers down the line, not just Gemini.
Large files are slow or expensive to analyze	Medium, hurts the experience for longer documents	Chunk long documents automatically, or set a soft length limit with a clear message to the user.
No accounts or persistent storage	Low to medium, users lose history when	Call this out clearly as a current limitation, and look at optional persistence in a later release.

Risk	Impact	Mitigation
	they switch sessions or devices	

13. Roadmap

Phase	What's in it	Target
Phase 1: Core (shipped)	Detection suite, highlight viewer, humanizer, PDF and DOCX ingestion, editorial toolkit, citation assistant	Complete
Phase 2: Reliability	Support for more than one AI provider, tighter schema validation, better performance on large files	Near-term
Phase 3: Persistence	Optional accounts with server-side history for documents and reports	Mid-term
Phase 4: Integrations	LMS integrations, a browser extension, and API access for outside tools	Long-term

14. Appendix

- **Repository:** github.com/SatyamPandey07/AUTHENTISCRIBE
- **License:** Apache-2.0
- **Runtime:** Node.js v18 or higher
- **Setup:** requires a GEMINI_API_KEY environment variable; PORT is optional and defaults to 3000.